Legality metrics

Before you celebrate wirelength, ask whether the placement is legal

The idea

- Check on-row placement, site alignment, in-chip bounds, and pairwise overlap
- Report displacement as L1 Manhattan distance from the origin layout
- Pair legality with HPWL after legalize

Pseudocode

- Before you trust wirelength, write a legality checker in pseudocode
- Inputs are coordinates and widths
- The loop walks cells and pairs
- The stop condition is the first failure reason, or ok when every check passes
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- The sketch also defines the metrics you report after a legalize pass
- On the overlap seed the checker fails with overlap A/B; the golden packing returns ok

Algorithm sketch — try these

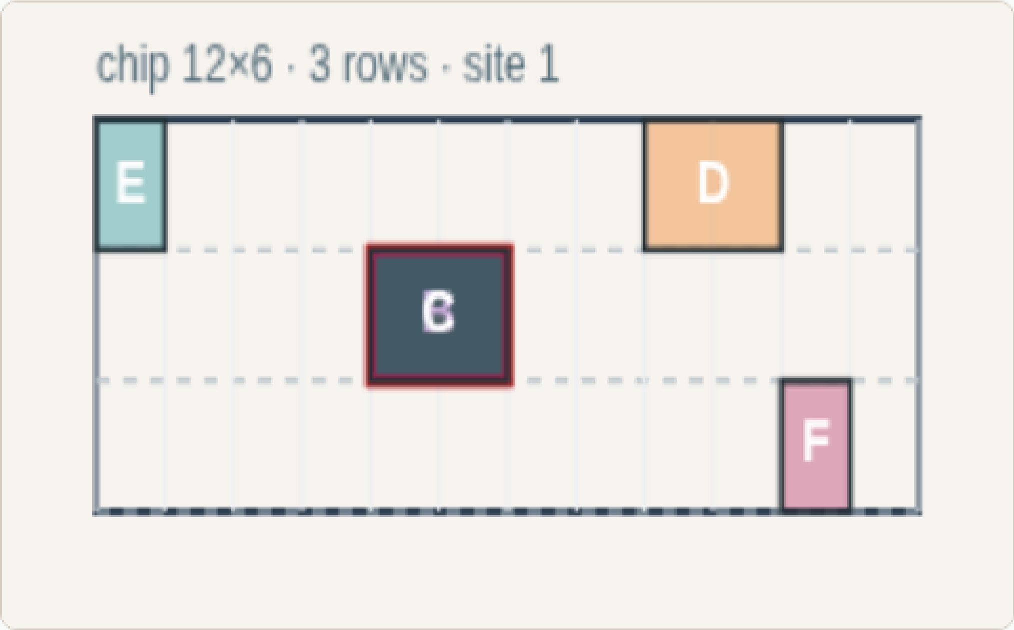
```
INPUT: positions, widths, chip, optional fixed
OUTPUT: legal?, reason, disp, HPWL
for each cell c:
  fail if off-site, off-row, outside, or macro moved
for each same-row pair (a,b):
  fail if [x,x+w) intervals overlap
disp = Σ|Δx|+|Δy|; HPWL = Σ net bbox (centers)
GOLDEN: overlap → "overlap A/B"; golden → "ok"
```

Overlap seed is illegal

LEGALITY METRICS

Overlap seed is illegal

Step 1 / 5 · overlap-illegal · module01-03-legality-metrics



Explanation

Starting from the overlap placement, the checker reports overlap A/B first—three cells share the middle row at x equals four. That single reason is enough to fail legality.

- Pairwise overlap on same row
- First reason wins the report
- Golden overlapIllegal: true

```
legal: false
reason: overlap A/B
all reasons: 3
```

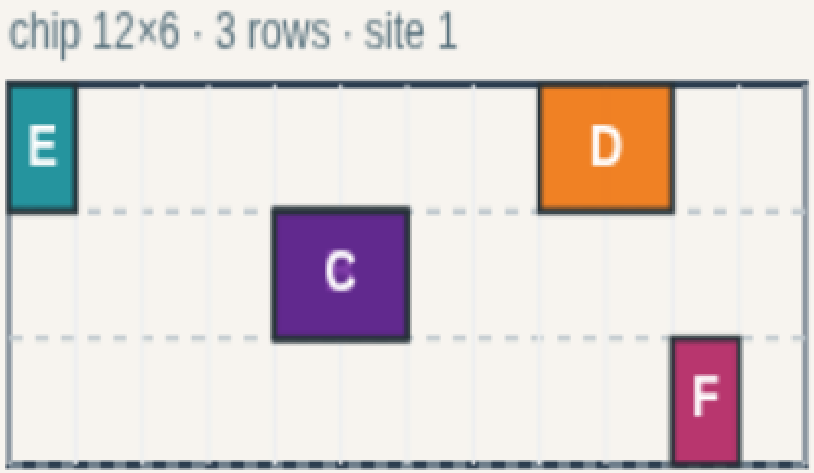
Overlap seed is illegal

Four legality checks

LEGALITY METRICS

Four legality checks

Step 2 / 5 · checks · module01-03-legality-metrics



Explanation

A legal placement must be on-row, site-aligned, inside the chip, and overlap-free. Fixed macros add a fifth check later—did the macro move off its lock?

- On-row: $y \in \{0, 2, 4\}$
- Site-aligned: integer x
- In-chip: $x + \text{width} \leq 12$
- No area overlap on a row

```
rows: 3
chip width: 12
edge-touch OK, area overlap not
```

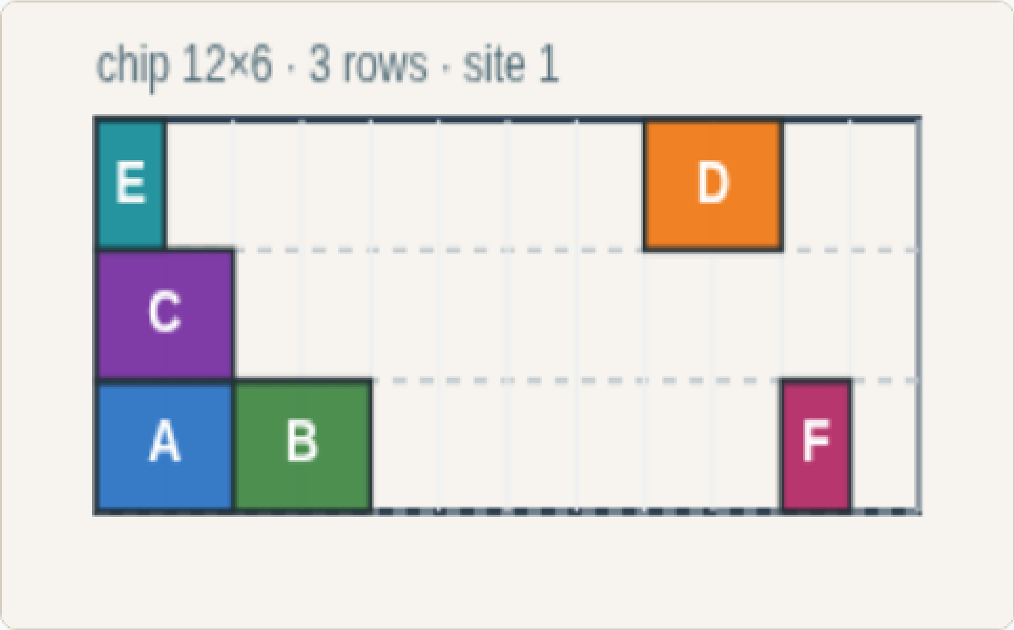
Four legality checks

Golden placement passes

LEGALITY METRICS

Golden placement passes

Step 3 / 5 · golden-ok · module01-03-legality-metrics



Explanation

The golden reference satisfies every check: reason ok, legal true. Use it as the positive control when you unit-test your legality reporter.

- All cells on valid rows
- Integer site coordinates
- No pairwise overlap

```
legal: true
reason: ok
goldenLegal: true
```

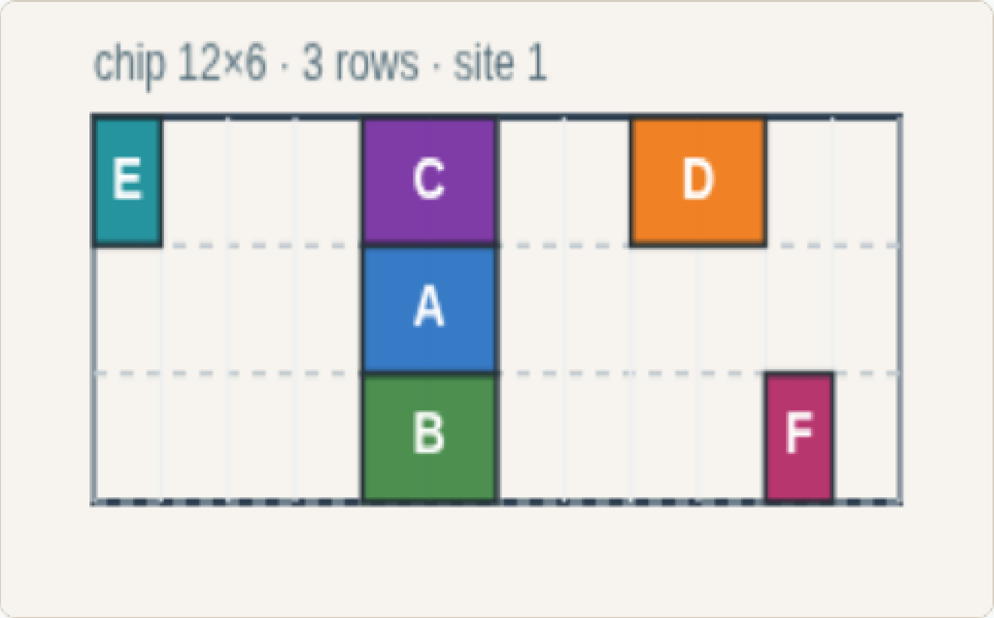
Golden placement passes

Displacement: L1 from origin

LEGALITY METRICS

Displacement: L1 from origin

Step 4 / 5 · displacement · module01-03-legality-metrics



Explanation

Displacement sums Manhattan distance per cell from a reference layout—here the overlap seed. Abacus later moves cells only four total units; overlap removal moves six.

- L1: $|\Delta x| + |\Delta y|$ per cell
- Sum over all movables
- Lower is closer to global place

```
overlap-removal disp: 6
abacus disp: 4
Report after legalize
```

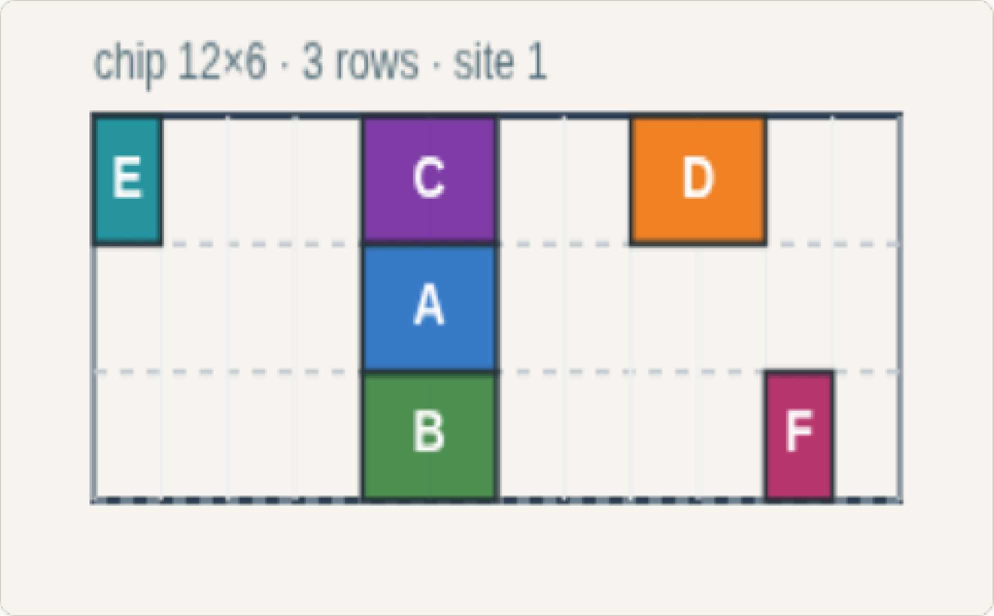
Displacement: L1 from origin

HPWL after legalize

LEGALITY METRICS

HPWL after legalize

Step 5 / 5 · hpwl-after · module01-03-legality-metrics



Explanation

Wirelength still matters: Abacus lands at HPWL thirty-eight with displacement four; Tetris-style packing hits HPWL thirty-two with displacement six. Always pair legality with both metrics.

- Same nets as placement course
- Cell centers for HPWL bbox
- Legal ≠ optimal HPWL

```
abacus HPWL: 38
tetris HPWL: 32
abacus disp: 4
```

HPWL after legalize

Browser lab track

- In the browser lab track, open the **legality-metrics** lab from the tools shelf
- Open the interactive lab
- Reveal golden is study-only
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module